# Exercise-1

**Aim**: Study of Linux general purpose utility commands and practice on file system / handling files.

**Hardware Requirements:** Processor type: RAM:

**Software Requirements:** Linux OS: Shell:

**Introduction:** Linux is a family of open-source Unix-like operating systems based on the Linux kernel.

Basic Features:

- Open Source − Linux source code is freely available and it is community based development project.
- Multi-User − Linux is a multiuser system means multiple users can access system resources like memory/ ram/ application programs at same time.
- Multiprogramming − Linux is a multiprogramming system means multiple applications can run at same time.
- Shell − Linux provides a special interpreter program which can be used to execute commands of the operating system.
- Security − Linux provides user security using authentication features like password protection/ controlled access to specific files/ encryption of data.

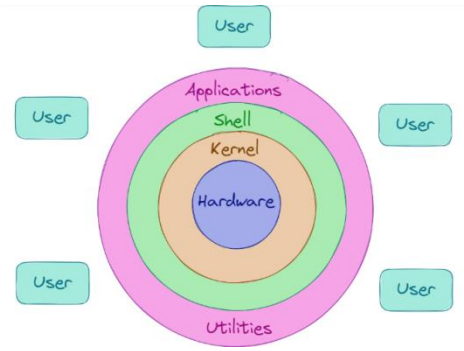



**Fig1.Linux Architecture**

Kernel: It is the primary component of OS and is typically in charge of all important OS tasks including process management, device management, etc.

**Types of shells:**

1. The Bourne Shell (sh)

Bourne shell is regarded as the first UNIX shell ever. It is denoted as sh. It gained popularity due to its compact nature and high speeds of operation. By default, it uses the prompt # for the root user and *$* for the non-root users.

2. The GNU Bourne-Again Shell (bash)

More popularly known as the Bash shell, the GNU Bourne-Again shell was designed to be compatible with the Bourne shell. It incorporates useful features from different types of shells in Linux such as Korn shell and C shell.

By default, it uses the prompt bash-VersionNumber# for the root user and bash-VersionNumber$ for the non-root users.

3. The C Shell (csh)

The C shell was created at the University of California by Bill Joy. It is denoted as csh. It was developed to include useful programming features like in-built support for arithmetic operations and a syntax similar to the C programming language.

By default, it uses the prompt *hostname#* for the root user and *hostname%* for the non-root users.

4. The Korn Shell (ksh)

To improve the Bourne shell. It is denoted as ksh. The Korn shell is essentially a superset of the Bourne shell.

By default, it uses the prompt # for the root user and *$* for the non-root users.

**Linux Distributions:**

Different parts of Linux are developed by different organizations. Different parts include kernel, shell utilities, X server, system environment, graphical programs, etc. Some of the popular Linux distros are as follows.

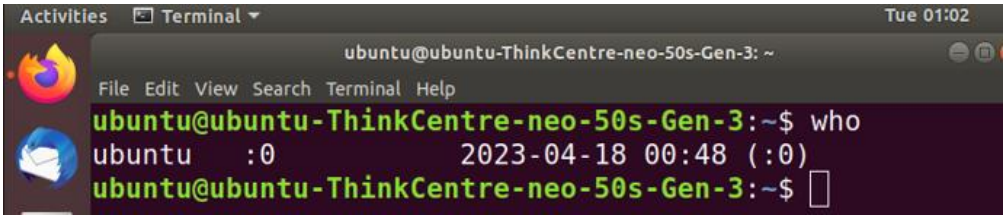
- Ubuntu Linux
- Red Hat Enterprise Linux
- Linux Mint
- Debian
- Fedora

**Procedure:**

Step1: Goto to Application App & launch the terminal window the logo is given below.



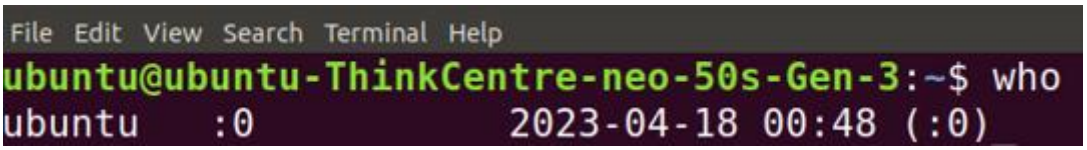
Step 2: It will open the shell that can be used to interpret the commands. The shell is shown in below.




**1.who** command can list the names of users currently logged in, their terminal, the time they have been logged in, and the name of the host from which they have logged in.

Syntax :- **who [options] [file]**

**Example1: $who**




| option | Meaning |
|---|---|
| **-H** | To print the heading of the columns displayed, |
| **-q** | To print the login names and total number of logged on users |
| **-m** | To show only hostname and user associated with stdin |

**2.echo,** It prints the given input string to standard output.

Syntax :- **echo string**

Example:- $ echo "welcome to Linux lab"
welcome to Linux lab

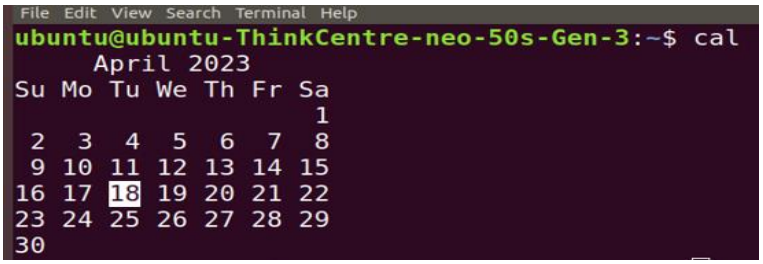
3**. cal**: Displays a calendar

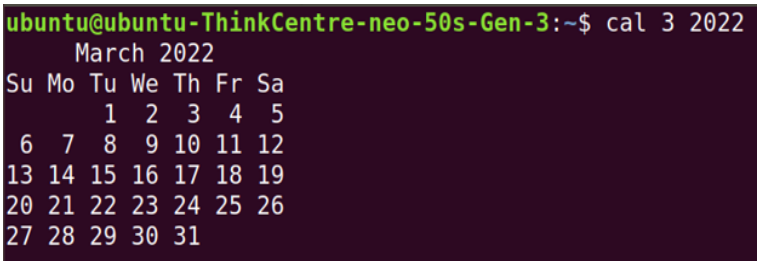
Syntax:- cal [options] [ month ] [year]

Description :- cal displays a simple calendar. If arguments are not specified, the current month is displayed. The switching options are as follows

| | |
|---|---|
| -1 | Display single (current) month output. (This is the default.) |
| -3 | Display prev/current/next month output |
| -s | Display Sunday as the first day of the week (This is the default.) |
| -m | Display Monday as the first day of the week |
| -j | Display Julian dates (days one-based, numbered from January 1) |
| -y | Display a calendar for the current year |

Example :




4. To find out the current shell we are working in

Syntax: $ **echo $0 or echo $SHELL**

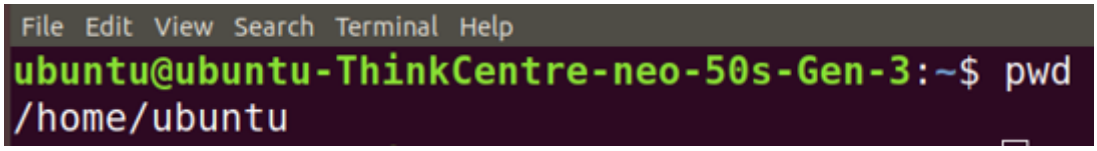
5. **clear** :- It clears the terminal screen.

Syntax :- $ **clear**

6. **pwd** :- Displays path from root to current directory

Syntax :- pwd

Example:



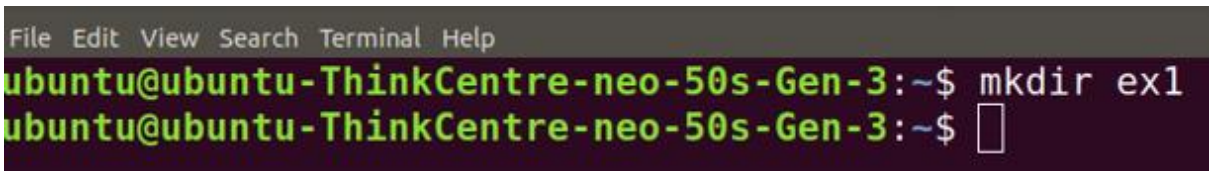

7. **mkdir:-** This command is used to create a new directory

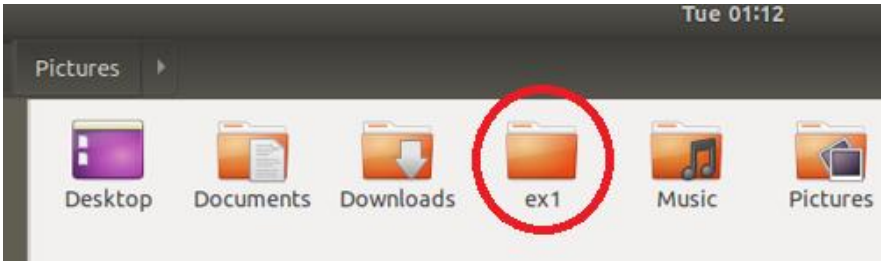
Syntax :- **mkdir [options] Directory**

Options**:**

| | |
|---|---|
| -m | Set permission mode (as in chmod) |
| -p | No error if existing, make parent directories as needed. |
| -v | Print a message for each created directory |
| Directory | The name of the directory that you wish to create |

Example:




Options Example: mkdir -m=rwx Directory name

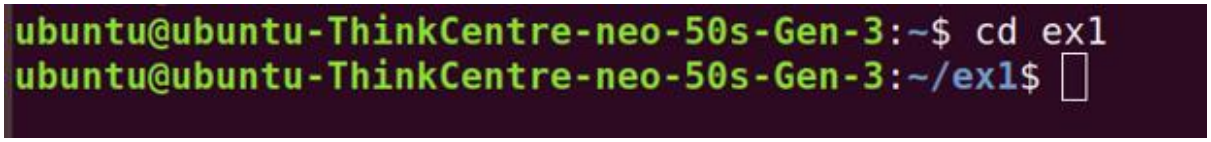
mkdir -p parent directory /subdirectory

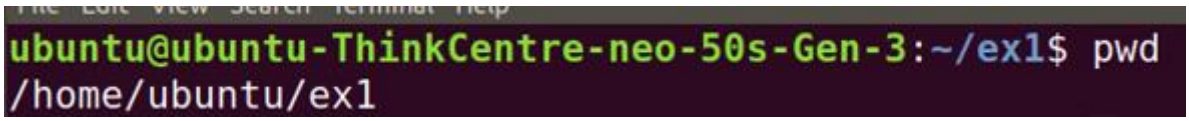
mkdir -v Ddirectory name

8. **cd** - It is used to change the directory.

Syntax :- cd [directory]

Example:



9. **cat:-** It is used to create, display and concatenate file contents.

Syntax : - cat [options] [FILE]...

| | |
|---|---|
| -A | Show all. |
| -b | Omits line numbers for blank space in the output. |
| -e | A $ character will be printed at the end of each line prior to a new line. |
| -E | Displays a $ (dollar sign) at the end of each line. |
| -n | Line numbers for all the output lines. |
| -s | If the output has multiple empty lines it replaces it with one empty line. |
| -T | Displays the tab characters in the output. |
| -v | Non-printing characters (with the exception of tabs, new-lines and form-feeds) are printed visibly. |

Uses of the cat command.

1) Create new files.

2) Display the contents of an existing file.

3) Concat

Create new files

Syntax: **cat > file1**

- Above syntax creates file1 and allow us to insert content for this file.
- After inserting content you can use ctrl+d to exit the file.
- If file with same name exist then it will overwrite that filenate the content of multiple files and display

Display the contents of an existing file

Syntax: **cat file1**

- Above syntax will display the content of file1

Concat

Syntax: **cat file1 >> file2**

- It can concatenate the contents of two files.
- For this you have to use append output redirection operator.
- The contents of file2 will be appended to file1

10. **cp**:- cp command copy files from one location to another. If the destination is an existing file, then the file is overwritten; if the destination is an existing directory, the file is copied into the directory (the directory is not overwritten).

Syntax :- **cp Source file destination file**

Example:- $ cp file1 file2

The above cp command copies the content of file1 to file2

11. **ls:-** Lists the contents of a directory

Syntax :- ls [options]

Options:

| | |
|---|---|
| -a | Shows you all files, even files that are hidden (these files begin with a dot.) |
| -A | List all files including the hidden files. However, does not display the working directory (.) or the parent directory (..). |
| -d | If an argument is a directory it only lists its name not its contents |
| -l | Shows you huge amounts of information (permissions, owners, size, and when last modified.) |
| -p | Displays a slash ( / ) in front of all directories |
| -r | Reverses the order of how the files are displayed |
| -R | Includes the contents of subdirectories |

Example:

```
ubuntu@ubuntu-ThinkCentre-neo-50s-Gen-3:~/ex1$ ls
ubuntu@ubuntu-ThinkCentre-neo-50s-Gen-3:~/ex1$
```

12. **rm**:- It is used to remove/delete the file from the directory.

Syntax :- **rm [options] [file|directory]**

Example: rm file1

- Remove the file myfile.txt. If the file is write-protected, you will be prompted to confirm that you really want to delete it.

Example: rm *

- Remove all files in the working directory. If it is write-protected, you will be

13. mv:- It is used to move/rename file from one directory to another.

Syntax :- **mv [options] old name  new name**

Example:

mv file1 file2
rename file1 to file2,  If the destination file doesn9t exist it will be created.

14. **touch:** creating a empty files.(Size of file would be zero)

Syntax: touch <file1>

15. wc – display the count of characters, words and lines of a file.

Syntax: wc  [option] [file]

| option | Meaning |
|---|---|
| -c | bytes |
| -m | characters |
| -l | No of lines |
| -w | No of words |

**Task:**

Log into Linux system and perform following operations on files.

1. Create a directory called  **your Roll No/ex1**
2. Change the directory to  **your Roll No/ex1**
3. Create Five empty files **empty1, empty2, empty3, empty4 and empty5**.(page 21-T2)
4. Create a file called **file1** and store your Roll No, Name, DOB and mail.
5. Display the contents of **file1** on screen.
6. Make a copy of the **file1** into another **file2.**
7. Combine the content of **file1** and **file2** into another **file3**
8. Delete the file **file1**
9. Display the all files including hidden files.
10. Count the no of lines and words of **file2**.
11. Compare the **file2** and **file3.**
12. Move the directory to new directory called **dump**.
13. What happens when you Enter the command ls  [aei]*.
14. What happens when you Enter the group command (wc file2;l –l file3) > file4
15. What happens when you Enter the command  **bc**

**for(1=1;1<=10;1++)**
**i**

**Conclusion:**